Alex Vance

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AI Systems & Platform Engineer — Agentic Systems, Deterministic Policy Gates, Scalable Distributed Infrastructure

EXPERIENCE

Synthetix Labs **San Francisco, CA**
Principal AI Infrastructure Engineer *2024 – Present*

- Architected multi-agent orchestration platform across 25+ production workflows serving enterprise workloads with automated multi-provider failover.
- Designed and implemented deterministic policy gates that validate model outputs before any external tool execution, ensuring fail-closed safety on uncertainty.
- Engineered Space Copilot, an advisory LLM-judge service evaluating production AI responses; redesigned its scoring metrics and resolved prompt-injection vulnerabilities.
- Built Mercury planner, turning natural language intent into codebase-grounded task trees bounded by strict verification gates before modifying central issue trackers.
- Led technical partnership evaluations and infrastructure integrations across leading foundation model inference providers.

CloudMatrix Systems **San Francisco, CA**
Senior Platform Engineer *2022 – 2024*

- Extracted 4 Go microservices from a legacy monolith with per-endpoint rollback safety and zero service disruption.
- Re-architected execution pipelines from synchronous HTTP blocking into an asynchronous queue-based job architecture with RabbitMQ and Redis, cutting latency from 60s to <100ms.
- Implemented automated payment and subscription infrastructure backed by 500+ unit and integration tests at 92%+ coverage behind a 5-stage CI/CD pipeline.
- Designed declarative pipeline orchestration reducing developer overhead for deploying new generation workflows by over 70%.
- Engineered multi-provider circuit breakers with priority-ordered fallback chains across external AI APIs.

DataCore Networks **San Jose, CA**
Distributed Systems Engineer *2020 – 2022*

- Designed high-throughput network data ingestion and telemetry pipelines processing millions of events daily.
- Optimized PostgreSQL database schemas and implemented multi-tier caching with Redis for measurable latency reductions.
- Maintained Docker-based infrastructure and automated testing pipelines for core platform microservices.

SELECTED OPEN SOURCE & RESEARCH PROJECTS

agent-gatekeeper • github.com/example-alex/agent-gatekeeper **MIT License**
Author & Maintainer *2024*

- Designed an open-source AI agent autonomy governance architecture featuring hard kill switches, token budgets, and live prompt-injection defense.
- Shipped end-to-end tooling across Go and TypeScript with widespread developer community adoption.

heimdall-cli • github.com/example-alex/heimdall-cli **MIT License**
Author & Maintainer *2023*

- Built a real-time meeting transcription and structured note synthesis CLI in Go with streaming LLM analysis.

EDUCATION

University of California, Berkeley **Berkeley, CA**
B.S. in Computer Science • GPA: 3.8 / 4.0 *2016 – 2020*

TECHNICAL SKILLS

- **Languages:** Go, Python, TypeScript, Node.js, SQL, C++, Bash
- **AI & Agent Tooling:** Model Context Protocol (MCP), Claude Code, OpenAI Function Calling, Agent Governance, RAG, Prompt Injection Guards
- **Infrastructure & Systems:** Docker, Kubernetes, PostgreSQL, Redis, RabbitMQ, Microservices, CI/CD, Distributed Systems, Linux
- **Practices:** Fail-Closed Policy Design, Deterministic Testing, Architecture Decision Records (ADRs)

CORE DESIGN THESIS

“AI proposes, deterministic code validates and gates, nothing with a real external effect happens without a hard policy check or human approval — fail closed on uncertainty, never fail open.”